ParT-EV: Edge-Conditioned Value Messages for Particle Transformer Jet Tagging

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October 5, 2026

1 Introduction

2 Related Work

3 Method

We study *ParT-EV*, a family of Particle Transformer extensions with edge-conditioned value messages. We use *EdgeValue* for the message mechanism and *ParT-EV* for models that include it. The main study retains ParT's standard kinematic pair features and compares EdgeValue under shared and layerwise attention bias.

3.1 Particle attention and pair information

Figure 1(a) shows the two input streams of ParT-EV. Particle features pass through an embedding with widths $(128, 512, 128)$ and eight particle self-attention blocks. In parallel, kinematic pair features are embedded and supplied to every particle-attention block. The attention width is 128 with eight heads, giving head dimension $d_h = 16$. Following the Particle Transformer (ParT) backbone [1], two class-attention blocks use the final particle representations to update a learned class token. A linear classifier and the final softmax convert this token into jet-class probabilities. EdgeValue changes the particle-attention computation, leaving this class readout unchanged. Our central question is whether using the existing pair information to condition value messages improves ParT, and whether that benefit depends on the choice of shared or layerwise attention bias.

For a valid query particle i and context particle j, the baseline supplies four kinematic pair features,

$$r_{ij} = \big(\Lambda(k_{t,ij}), \Lambda(\zeta_{ij}), \Lambda(\Delta_{ij}), \Lambda(m^2_{ij})\big), \tag{1}$$

where $\Lambda(x) = \log(\max\{x, 10^{-8}\})$. Here Δ is separation in rapidity–azimuth space, k_t the relative transverse-momentum scale, ζ the smaller transverse-momentum share, and m^2 the pair invariant mass squared. The baseline therefore already incorporates pairwise kinematics within global attention.

The upper path in Figure 1(b) is the usual ParT attention computation: a pair-derived bias is added to the query–key score before a masked softmax determines the attention weights.

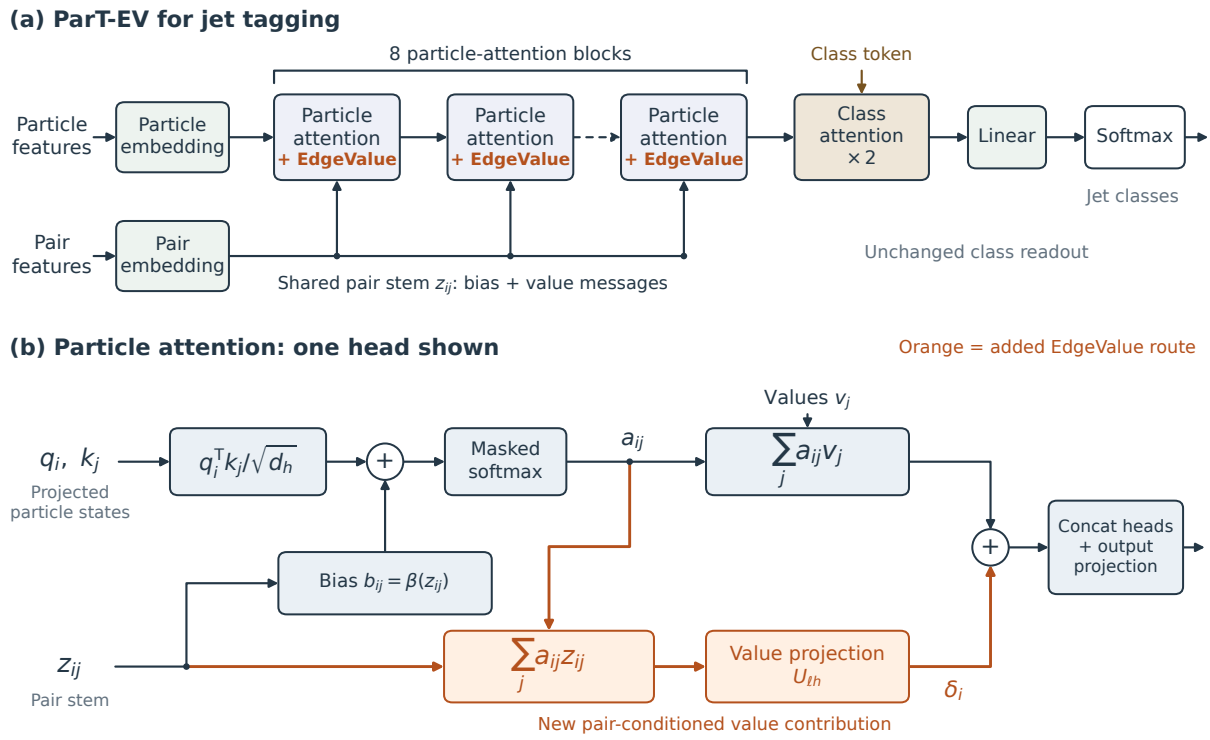


Figure 1: ParT-EV architecture. (a) Particle and pair features follow parallel embedding streams; the pair stem is supplied to each of the eight particle-attention blocks. The class-attention readout and classifier retain the ParT design [1]. (b) One attention head: the usual pair-biased attention path is shown in dark blue. The added EdgeValue route (orange) aggregates pair embeddings with the same attention weights, projects the result with $U_{\ell h}$, and adds it to the ordinary value sum before head concatenation and output projection. Surrounding normalization, residual, and feed-forward operations are omitted. Evaluation-time weights are shown; layer/head indices are suppressed except on $U_{\ell h}$. The bias map may be shared or layerwise.

For layer ℓ and head h, biased attention is [1, 2]

$$
\begin{aligned}
L_{ij}^{\ell h} &= \frac{(q_i^{\ell h})^\top k_j^{\ell h}}{\sqrt{d_h}} + b_{ij}^{\ell h}, \\
a_{ij}^{\ell h} &= \frac{\exp L_{ij}^{\ell h}}{\sum_{k \in \mathcal{V}} \exp L_{ik}^{\ell h}}, \qquad i, j \in \mathcal{V}.
\end{aligned}
\tag{2}
$$

Here $\mathcal{V}$ is the set of valid constituents, so the normalization excludes padded keys and permits self-attention ($j = i$). We write $\widetilde{a} = \mathrm{Dropout}(a)$ for the attention weights after training-time dropout; at evaluation $\widetilde{a} = a$. Bias-only attention aggregates $\sum_{j \in \mathcal{V}} \widetilde{a}_{ij}^{\ell h} v_j^{\ell h}$: relations affect the weights, while each context value remains independent of its query.

3.2 Shared and layerwise interaction biases

A three-stage, width-64 pair network maps the standard pair features to a common relation stem $z_{ij} = \phi(r_{ij}) \in \mathbb{R}^{64}$, corresponding to the lower embedding stream in Figure 1(a). The bias branch in

panel (b) converts this representation into one scalar bias per attention head. Shared bias uses one head-output map β in all particle layers, $b_{ij}^{\ell h} = \beta_h(z_{ij})$. Layerwise bias uses independently trainable maps β_ℓ, so $b_{ij}^{\ell h} = \beta_{\ell h}(z_{ij})$, while retaining the common stem. The layerwise maps, including their output normalization, have identical initial parameters.

3.3 Relation-conditioned value messages

Pairwise attention bias influences how strongly a query receives a context particle's value. In bias-only ParT, however, the underlying value vector $v_j^{\ell h}$ is the same for every query receiving information from j within that layer and head. EdgeValue also makes the message content depend on the relation between the two particles.

The orange route in Figure 1(b) gives the pair embedding a second role: in addition to supplying an attention bias, it contributes a learned vector to the value message. The extra vector for pair (i, j) is $U_{\ell h} z_{ij}$, and it receives the same attention weight as the ordinary value $v_j^{\ell h}$. Building on relation-aware value messages [3], ParT-EV therefore uses

$$y_i^{\ell h} = \sum_{j \in \mathcal{V}} \widetilde{a}_{ij}^{\ell h} \left(v_j^{\ell h} + U_{\ell h} z_{ij} \right), \tag{3}$$

$$\delta_i^{\ell h} = \sum_{j \in \mathcal{V}} \widetilde{a}_{ij}^{\ell h} U_{\ell h} z_{ij}, \qquad U_{\ell h} \in \mathbb{R}^{16 \times 64}. \tag{4}$$

Here $y_i^{\ell h}$ is the head aggregate before the existing output projection, and $\delta_i^{\ell h}$ is the addition to the ordinary weighted value sum. Both terms use the same attention weights $\widetilde{a}$ and the same training-time attention-dropout realization.

The figure shows an equivalent, efficient ordering of this computation. Because $U_{\ell h}$ is linear and independent of i and j, the pair embeddings can be aggregated before projection:

$$\delta_i^{\ell h} = U_{\ell h} \left(\sum_{j \in \mathcal{V}} \widetilde{a}_{ij}^{\ell h} z_{ij} \right). \tag{5}$$

The first orange block computes the attention-weighted sum of pair embeddings; the second maps it from 64 channels to the 16-dimensional head value space. At the addition node on the right, this contribution joins the ordinary weighted value sum. Head outputs are then concatenated and passed through the existing output projection. This ordering avoids materializing a separate pair-value vector for every head, although pair features and attention weights still require quadratic storage.

The pair stem is encoded from the input descriptors and shared across particle-attention layers, rather than updated from their evolving particle states. Its encoder is trained jointly with the classifier. Each layer and head has its own projection $U_{\ell h}$, so the same pair representation can contribute differently at different depths and through different heads. The orange route is present under either bias scheme: EdgeValue requires neither layerwise bias nor additional pair descriptors.

The value-message projections are bias-free. Within each layer, their stacked $8 \times 16 \times 64$ parameter tensor is initialized with Xavier-uniform initialization and optimized jointly with the rest of the network. Setting all $U_{\ell h} = 0$ removes the additional term and recovers the corresponding bias-only attention operation for fixed remaining parameters. Invalid pair stems and padded-query edge contributions are zeroed. The surrounding normalization, residual, and feed-forward operations remain unchanged. Applied to all eight particle-attention blocks in panel (a), the projections add $8 \times 8 \times 16 \times 64 = 65{,}536$ trainable parameters relative to the matching bias-only path.

4 Experimental Setup

4.1 Data and study design

We study ten-class jet tagging using the offline constituent view of the JetClass dataset [4], introduced with Particle Transformer (ParT) [1]. The classes are QCD jets and nine signal categories: $H \to b\bar{b}$, $H \to c\bar{c}$, $H \to gg$, $H \to 4q$, $H \to q\bar{q}'\ell\nu$, $Z \to q\bar{q}$, $W \to q\bar{q}'$, fully hadronic top decay ($t \to bqq'$), and leptonic top decay ($t \to b\ell\nu$). All models receive the same reconstructed offline information.

The training population contains 1,000,000 jets, balanced across the ten classes. A disjoint, likewise balanced validation population contains 125,000 jets and controls stopping and checkpoint selection. These populations are sampled without replacement from a 10M-event portion of the training release. Evaluation uses the official *20-million-jet JetClass test sample*: 2,000,000 jets per class, including 2,000,000 QCD jets, disjoint from the training and validation samples.

The main study compares four configurations, crossing shared versus layerwise attention bias with the absence versus presence of EdgeValue messages. All four retain the standard ParT pair inputs. We refer to models without EdgeValue as ParT and those with it as ParT-EV, specifying shared or layerwise bias where needed. The baseline is ParT with shared bias. Every configuration is trained with three random seeds, matched across configurations, with the same training-event order for a given seed and epoch. Headline gains are measured relative to this shared-bias ParT baseline trained under the same protocol. Ablation comparisons instead hold the other factors fixed: for example, the EdgeValue gain compares models with the same pair features and bias scheme. Each table specifies its reference configuration. A supplementary extension to richer pair representations, including its methods and experimental design, is presented in Appendix A.

4.2 Inputs and training

Each jet is represented by at most 128 constituents in their original order, with padding masked. The common 17-dimensional particle input contains log transverse momentum and energy, their log fractions relative to the jet, radial and signed angular offsets, charge, five particle-identification indicators, and transverse/longitudinal impact parameters with their uncertainties. Jet-relative quantities are calculated from the constituent four-vectors. Apart from the bias scheme and the addition of EdgeValue, the backbone architecture in Section 3 is held fixed.

All models are trained from scratch with unweighted ten-class cross-entropy and AdamW [5], using $(\beta_1, \beta_2) = (0.9, 0.999)$, weight decay 10^{-4}, and a gradient-norm clipping threshold of 1. Gradients are accumulated over two microbatches of 64 jets, giving an effective batch size of 128. Training uses BF16 mixed precision; test inference uses FP32.

The training budget is 40 epochs, with a minimum of 12 epochs and an early-stopping patience of eight epochs. The learning rate increases linearly to 10^{-3} and then follows a cosine decay toward 10^{-5}. Warm-up spans the first 5% of the planned optimizer updates; the schedule is fixed to the full 40-epoch budget regardless of early stopping.

Checkpoints are selected by validation accuracy, using lower cross-entropy to distinguish epochs within 0.01 percentage points of the best accuracy. The same selection criterion governs early stopping for every configuration.

The comparison matches data, backbone dimensions, loss, optimizer, and training budget, but does not impose equal parameter counts or equal training time. Shared-bias models use sequence trimming, whereas layerwise-bias models do not; their pair-processing paths also yield different BatchNorm populations. Shared-versus-layerwise differences therefore cannot be attributed solely

to bias sharing. We assess EdgeValue by comparing models with the same feature set and bias scheme, which retain the same trimming and pair-processing choices.

4.3 Classification and background-rejection metrics

We summarize classification performance with multiclass accuracy and macro one-versus-rest area under the ROC curve (macro AUROC) [6]. For each trained model, class AUCs rank events by the corresponding softmax probability; macro AUROC is their unweighted mean over the ten classes. We report arithmetic means and sample standard deviations across three independently trained models, rather than combining their predictions into an ensemble. Accuracy and macro-AUROC differences from baseline are expressed in percentage points.

Background rejection is measured separately for each signal class s against QCD. If $\ell_c(x)$ denotes the classifier logit for class c, we use

$$\begin{aligned} D_s(x) &= \ell_s(x) - \ell_{\mathrm{QCD}}(x), \\ \widehat{\epsilon}_{\mathrm{QCD},s}(t) &= \frac{1}{n_{\mathrm{QCD}}} \sum_{x \in \mathcal{Q}} \mathbf{1}[D_s(x) \geq t]\,, \\ R_s(\epsilon_s) &= \frac{1}{\widehat{\epsilon}_{\mathrm{QCD},s}(t_s)}, \qquad \widehat{\epsilon}_{\mathrm{QCD},s}(t_s) > 0. \end{aligned} \tag{6}$$

Here $\mathcal{Q}$ is the set of test QCD jets, $n_{\mathrm{QCD}} = |\mathcal{Q}|$, and $\mathbf{1}[\cdot]$ is the indicator function; t_s is the threshold at target signal efficiency ϵ_s, defined below. The main rejection table follows the JetClass benchmark operating points: 50% signal efficiency for the seven hadronic signal classes, 99% for $H \to q\bar{q}'\ell\nu$, and 99.5% for $t \to b\ell\nu$ [1]. Each row states its signal efficiency; the class-specific choices are the same for every configuration and seed. The main table compares all four standard-feature configurations, pairing the models with and without EdgeValue under each bias scheme. Appendix B provides the complete numerical results, including the richer-representation extension and rejection at 30% efficiency for the seven hadronic classes. All operating points use the same checkpoints and test predictions.

The discriminant is monotonic in $p_s/(p_s + p_{\mathrm{QCD}})$. At target efficiency ϵ_s, t_s is the $\lceil \epsilon_s n_s \rceil$-th largest score among the n_s true signal events. All events with $D_s \geq t_s$ pass, including ties; the achieved signal efficiency can therefore exceed the target because of integer rounding or tied scores. Thresholds are determined separately for each signal and trained model from the full test sample.

5 Results

We first compare standard shared-bias ParT with and without edge-conditioned value messages, then examine the bias scheme and class-wise background rejection. The main comparisons retain standard ParT pair features. All results use the same official 20M-jet test sample and are averaged over three independently trained models per configuration. Reported spreads are sample standard deviations across training seeds, not confidence intervals or ensemble results.

5.1 EdgeValue with standard shared bias

Table 1 compares the baseline with its direct EdgeValue extension, retaining the standard four pair features and shared attention bias. Accuracy increases from $82.819 \pm 0.100\%$ to $83.474 \pm 0.047\%$, a gain of 0.655 percentage points (pp). Macro AUROC increases from 98.229% to 98.342%, a gain of 0.113 pp. The accuracy improvement is positive for each matched seed, ranging from 0.597 to 0.726 pp.

Table 1: Adding EdgeValue to standard ParT with shared pair bias. Both models use the standard pair features and are trained on 1M jets; evaluation uses the same official 20M-jet test sample. Values are means ± sample standard deviations across three training seeds. Δ denotes the difference from ParT in percentage points.

Model	Accuracy (%)	Δ (pp)	Macro AUROC (%)	Δ (pp)
ParT	82.819 ± 0.100	0.000	98.229 ± 0.018	0.000
ParT-EV	83.474 ± 0.047	+0.655	98.342 ± 0.013	+0.113

This comparison demonstrates that additional pair descriptors and layerwise bias are not required to obtain an improvement in this setting. The existing kinematic pair representation becomes more useful when it can contribute to the vector content of messages as well as their attention weights. The comparison holds the inputs and bias scheme fixed, but adds the EdgeValue projections described in Section 3.3; it is not a parameter-matched test against a larger bias-only model.

5.2 Shared versus layerwise bias

Table 2 compares the four standard-feature configurations under shared and layerwise bias. Each row compares otherwise matched models with and without EdgeValue; comparisons between rows measure differences associated with the bias scheme and its implementation.

With standard pair features, moving from shared to layerwise bias changes the mean accuracy by only +0.040 pp without EdgeValue. With EdgeValue, the layerwise configuration reaches 83.438%, slightly below the shared-bias model's 83.474% (−0.036 pp). Neither ordering holds for all three seeds: layerwise bias improves accuracy in two seeds without EdgeValue and in one seed with EdgeValue. These small mean differences do not show a consistent advantage for the layerwise configuration with standard inputs.

By contrast, adding EdgeValue improves accuracy under both bias schemes: +0.655 pp with shared bias and +0.579 pp with layerwise bias. Both comparisons improve in all three seeds. The benefit of edge-conditioned messages therefore persists across these two implementations rather than depending on the layerwise bias choice. As discussed in Section 4.2, shared-versus-layerwise comparisons also change trimming and pair-normalization handling, so they should not be interpreted as isolating bias sharing alone.

Table 2: EdgeValue with shared and layerwise pair bias on the official 20M-jet test sample after training on 1M jets. All four configurations use the standard ParT pair features. Within each row, only EdgeValue is switched on or off. Values are means ± sample standard deviations over three seeds; EV gain is the difference of the two means in percentage points.

(a) Accuracy (%)

Pair bias	Without EV	With EV	EV gain (pp)
Shared	82.819 ± 0.100	83.474 ± 0.047	+0.655
Layerwise	82.859 ± 0.055	83.438 ± 0.052	+0.579

(b) Macro AUROC (%)

Pair bias	Without EV	With EV	EV gain (pp)
Shared	98.229 ± 0.018	98.342 ± 0.013	+0.113
Layerwise	98.233 ± 0.008	98.330 ± 0.006	+0.096

Table 3: QCD background rejection on the official 20M-jet test sample at the JetClass benchmark signal efficiencies [1]. The first two model columns compare standard shared-bias ParT without and with EV; the last two compare layerwise models without and with EV. All models use standard ParT pair features and are trained on 1M jets. Values are means ± sample standard deviations over three seeds; higher is better.

		Shared bias		Layerwise bias	
Signal	ϵ_s (%)	ParT	ParT-EV	ParT	ParT-EV
$H \to b\bar{b}$	50	5271 ± 242	5674 ± 152	5191 ± 195	5408 ± 248
$H \to c\bar{c}$	50	1741 ± 31	2007 ± 15	1792 ± 84	2020 ± 143
$H \to gg$	50	90.3 ± 0.8	94.1 ± 0.8	91.2 ± 0.7	93.7 ± 1.7
$H \to 4q$	50	665.0 ± 22.3	813.5 ± 10.0	683.6 ± 32.5	788.6 ± 24.0
$H \to q\bar{q}'\ell\nu$	99	1564 ± 23	1832 ± 32	1535 ± 126	1855 ± 106
$Z \to q\bar{q}$	50	229.8 ± 3.2	249.5 ± 4.9	234.8 ± 5.0	249.1 ± 4.4
$W \to q\bar{q}'$	50	284.7 ± 7.8	315.2 ± 9.4	292.1 ± 2.7	311.7 ± 2.3
$t \to bq\bar{q}'$	50	5833 ± 464	7395 ± 263	5629 ± 40	6932 ± 761
$t \to b\ell\nu$	99.5	3903 ± 161	4207 ± 182	3541 ± 348	4267 ± 378

5.3 Class-wise background rejection

Table 3 compares the four standard-feature configurations at the JetClass benchmark operating points. The columns pair each ParT-EV variant with its corresponding bias-only control, retaining the same pair inputs and bias scheme within each comparison. Complete numerical results are provided in Appendix B.

The standard shared-bias EdgeValue extension increases the mean background rejection for all nine signal classes at these operating points. Among the hadronic classes at 50% signal efficiency, the relative increases range from 4.2% for $H \to gg$ to 26.8% for fully hadronic top jets. For $H \to 4q$, rejection rises from 665.0 to 813.5 (+22.3%), while for $H \to b\bar{b}$ it rises from 5271 to 5674 (+7.6%). Thus, the improvement observed in multiclass accuracy is accompanied by better signal-versus-QCD discrimination across the benchmark categories.

With layerwise bias, adding EdgeValue likewise improves mean benchmark rejection in all nine classes relative to its matching bias-only control. The gains include 15.4% for $H \to 4q$ and 23.2% for fully hadronic top jets at 50% efficiency. Thus, the class-wise improvements also persist under

both bias schemes, although their sizes differ by class.

At the stricter 30% operating point (Table 6), the standard shared-bias EdgeValue model improves mean rejection over baseline in all seven hadronic classes, including 28.1% for $H \to 4q$ and 35.6% for fully hadronic top jets. These comparisons show that the benefits extend to a lower-efficiency regime. All rejection gains quoted here are relative changes in mean rejection, distinct from the percentage-point changes used for accuracy and macro AUROC; point-estimate differences alone do not establish statistical significance.

Beyond standard pair features. Appendix A examines richer physics-inspired pair representations and finds positive EdgeValue accuracy gains under both bias schemes. These supplementary results motivate a broader study of which relations best complement edge-conditioned value messages, without requiring those additional descriptors for the gains demonstrated in the main study.

6 Conclusion

References

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A Extension to richer pair representations

The main study uses only the four standard ParT pair features. Here we extend that representation with three physics-inspired relation families and examine whether the EdgeValue gains persist. This supplementary study varies the pair representation while retaining the attention and value-message mechanisms described in Section 3.

A.1 Additional relation descriptors

For the comparison in this appendix, we denote the main study's four-feature pair input r_{ij} by r_{ij}^{std}. The selected input concatenates these standard channels with learned PT, TRACK, and REGION embeddings of dimensions 8, 12, and 12, respectively, giving $r_{ij}^{\mathrm{sel}} \in \mathbb{R}^{36}$. The common width-64 stem now receives r_{ij}^{sel} instead of r_{ij}^{std}. Throughout, i denotes the query and j the context, and directional features are not symmetrized. TRACK and REGION use pseudorapidity for angular separations, whereas the standard ParT pair features use rapidity.

PT. Let $S_T = \sum_{k\in\mathcal{V}} p_{T,k}$ be the scalar constituent-p_T sum and $\epsilon = 10^{-6}$ the numerical safeguard used by the added relation families. Each endpoint supplies $p_{T,i}/(S_T+\epsilon)$, the safeguarded log-fraction $\log[(p_{T,i}+\epsilon)/(S_T+\epsilon)]$, and its descending normalized p_T rank. The remaining four channels are the context-minus-query log-fraction and rank differences, the pair log-fraction $\log[(p_{T,i}+p_{T,j})/(S_T+\epsilon)+\epsilon]$, and $(p_{T,j}-p_{T,i})/(p_{T,j}+p_{T,i}+\epsilon)$. Tied ranks are averaged. A $10\to32\to8$ network embeds these ten quantities.

TRACK. Each endpoint supplies d_0, d_z, logarithmic effective uncertainties, inverse-hyperbolic-sine impact-parameter significances, and a validity flag. Validity requires charged-hadron, electron, or muon identification, finite, nonmissing measurements, and positive uncertainties. Effective errors satisfy $\sigma_{u,\mathrm{eff}}^2 = \sigma_u^2 + f_u^2$ for $u \in \{d_0, d_z\}$, with floors f_u fitted on training data. Shared endpoint weights produce $g_i, g_j \in \mathbb{R}^{16}$ using a $7\to32\to16$ network. The pair encoder receives

$$[g_i, g_j, g_j - g_i, g_i \odot g_j, s_{ij}, c_{ij}] \in \mathbb{R}^{81}, \tag{7}$$

where s_{ij} is a four-component one-hot encoding of the ordered endpoint validity states and c_{ij} contains 13 compatibility channels. These comprise uncertainty-normalized context-minus-query displacement differences, their squared-sum χ^2 through $\log(1+\chi^2)$ and $\exp[-\min(\chi^2, 25)/2]$, extrema of absolute transformed significances and their signed products, and angular descriptors $\sin\Delta\phi$, $\cos\Delta\phi$, and $\log(\Delta R+\epsilon)$, with $\Delta\phi = \phi_i - \phi_j$ wrapped periodically and ΔR measured in pseudorapidity–azimuth space. Compatibility channels vanish unless both tracks are valid. The resulting $81\to48\to12$ network retains missing-measurement information through the validity states.

REGION. A deterministic, beam-free angular tree repeatedly merges the closest pair in (η,ϕ) using four-vector addition. It follows the angular-clustering motivation of Cambridge/Aachen [7, 8], with the explicit pseudorapidity-based prescription above. Exclusive partitions at $K = 2, 4, 8$ clusters, capped by constituent multiplicity, provide 41 features: three same-cluster indicators; five lowest-common-ancestor descriptors (normalized depth, merge angle, k_t, momentum sharing, and mass); 18 endpoint cluster p_T, mass, and multiplicity fractions; six constituent shares within those clusters; six constituent-to-cluster-axis distances; and three context-minus-query cluster-p_T rank differences. Angles use reference radius $R_{\mathrm{ref}} = 0.8$, merge k_t uses $p_{T,\mathrm{jet}}R_{\mathrm{ref}}$, and mass uses jet

mass. Cluster p_T and mass fractions, and merge angle, k_t, and mass descriptors use logarithms. Undefined diagonal merge quantities are zeroed. A $41 \to 32 \to 12$ network embeds the result. These descriptors are supplied for every valid pair, while attention remains global.

Here jet p_T and mass are obtained from the summed constituent four-vector, not the scalar sum S_T used by PT. Ancestor depth is divided by the maximum leaf depth, with the denominator bounded below by one.

A.2 Preprocessing and experimental design

All family networks use GELU and RMS normalization between their two linear maps and are trained jointly with the classifier. Before entering these networks, designated continuous features are centered and rescaled using statistics from a fixed subset of 50,000 training jets. This subset is used only to estimate preprocessing constants; the classifier is trained on the full one-million-jet training sample. For each feature, we subtract its median, divide by a robust measure of its spread, and clip extreme values:

$$\widehat{x} = \mathrm{clip}_{[-8,8]} \left(\frac{x - \mathrm{median}(x)}{\max[(Q_{75} - Q_{25})/1.349, 10^{-6}]} \right). \tag{8}$$

Here Q_{25} and Q_{75} are the training-sample quartiles; the resulting scale is less sensitive to outliers than a standard deviation. Track uncertainty floors are also estimated from training data. Binary and designated fixed-scale features are left unchanged. These preprocessing constants remain fixed during validation and testing and are shared across the selected-input configurations. Padded pair embeddings are zeroed.

The supplementary study repeats the shared/layerwise bias and EdgeValue off/on comparisons with the richer representation, giving four additional configurations. For the appendix comparisons, `Std` denotes the standard pair inputs used in the main study and `Sel` denotes the richer input. The following `S` or `L` denotes shared or layerwise bias, and `+EV` indicates EdgeValue. Thus, `Std-S` is the main-study ParT baseline, while `Sel-L+EV` uses the added relations, layerwise bias, and EdgeValue. All four additional configurations use the same data splits, three matched random seeds, training protocol, and checkpoint-selection rule as the main study. The selected relations use the same constituents and track measurements, without additional detector information. Their encoders add trainable parameters, so comparisons with standard inputs are not capacity matched. The complete eight-configuration results appear in Appendix B.

A.3 Results with richer pair representations

Table 4 includes the four selected-input configurations. At fixed bias scheme and EdgeValue setting, adding these relations improves mean accuracy in every comparison. The gains are 0.400 pp for shared bias without EdgeValue and 0.130 pp for shared bias with EdgeValue. For layerwise bias, the corresponding gains are 0.654 and 0.409 pp. All four selected-versus-standard comparisons improve accuracy in each of the three matched seeds.

The bias-scheme comparison also changes with the richer relation input: the selected-relation layerwise configuration exceeds its shared counterpart by 0.294 pp without EdgeValue and 0.243 pp with EdgeValue, consistently across seeds. Additional relation information thus has a larger measured benefit in the layerwise implementations. This is an empirical comparison of the complete configurations, subject to the trimming and pair-processing differences noted in Section 4.2, rather than a causal attribution to separate layerwise bias maps.

EdgeValue remains beneficial after adding the selected relations, contributing 0.385 pp with shared bias and 0.334 pp with layerwise bias, again with positive gains in all three seeds. The improvements are not simply additive: the marginal EdgeValue gain is smaller with the selected features than with the standard features. Nevertheless, the selected-relation layerwise ParT-EV achieves the highest mean accuracy and macro AUROC among the eight configurations: $83.847 \pm 0.042\%$ and $98.399 \pm 0.014\%$, respectively. Relative to baseline, these are improvements of 1.028 pp in accuracy and 0.170 pp in macro AUROC. These total gains combine the change in pair representation, bias scheme, and value messages; they do not measure EdgeValue alone.

Class-wise gains are not uniform (Table 5). Relative to baseline, selected-relation layerwise ParT-EV increases mean rejection at 50% signal efficiency by 28.3% for $H \to c\bar{c}$, 30.8% for $H \to 4q$, and 37.1% for fully hadronic top jets. At the leptonic benchmark efficiencies, the gains are 23.5% for $H \to q\bar{q}'\ell\nu$ and 21.9% for $t \to b\ell\nu$. The matched comparison with the selected-relation layerwise model without EdgeValue improves mean rejection in eight of the nine classes. The exception is $H \to b\bar{b}$: rejection is 5253, compared with 5271 for baseline, and below the standard shared-bias EdgeValue model. The small -0.3% difference from baseline does not by itself establish a degradation. At 30% efficiency (Table 6), gains over baseline include 42.9% for $H \to 4q$ and 56.2% for fully hadronic top jets.

These results show that EdgeValue can also exploit a richer pair representation, but do not identify an optimal set of relations. PT, TRACK, and REGION are added together, so their individual contributions and interactions remain unresolved. A broader study of which physics-inspired relations most benefit ParT-EV, including individual-family ablations and capacity-matched controls, is an important direction for future work.

B Complete numerical results

Table 4 lists accuracy and macro AUROC for all eight configurations, with differences measured against the shared-bias standard ParT baseline. Tables 5 and 6 give the complete class-wise rejection results at the benchmark efficiencies and at 30% signal efficiency, respectively. The appendix configuration labels are defined in Appendix A.2: `Std` uses standard pair inputs, `Sel` adds PT, TRACK, and REGION, `S/L` identifies shared/layerwise bias, and `+EV` denotes edge-conditioned value messages. Every configuration uses three independently trained models evaluated on the same 20M test jets; the reported spreads are sample standard deviations across seeds.

Table 4: Offline classification on the official 20M-jet test sample for models trained on 1M jets. Values are means ± sample standard deviations over three seeds; Δ is the difference from baseline `Std-S` in percentage points. Std and Sel denote standard and standard + PT, TRACK, REGION pair inputs; S and L denote shared and layerwise bias.

ID	Accuracy (%)	Δ (pp)	Macro AUROC (%)	Δ (pp)
`Std-S`	82.819 ± 0.100	0.000	98.229 ± 0.018	0.000
`Std-L`	82.859 ± 0.055	+0.040	98.233 ± 0.008	+0.004
`Std-S+EV`	83.474 ± 0.047	+0.655	98.342 ± 0.013	+0.113
`Std-L+EV`	83.438 ± 0.052	+0.619	98.330 ± 0.006	+0.101
`Sel-S`	83.219 ± 0.061	+0.400	98.294 ± 0.016	+0.065
`Sel-L`	83.513 ± 0.092	+0.694	98.343 ± 0.012	+0.114
`Sel-S+EV`	83.604 ± 0.023	+0.785	98.364 ± 0.007	+0.134
`Sel-L+EV`	83.847 ± 0.042	+1.028	98.399 ± 0.014	+0.170

Table 5: QCD background rejection on the official 20M-jet test sample at the JetClass benchmark signal efficiencies [1]: 50% for the seven hadronic classes, 99% for $H \to q\bar{q}'\ell\nu$, and 99.5% for $t \to b\ell\nu$. Panels compare standard and selected pair inputs, with all models trained on 1M jets. Values are means ± sample standard deviations over three seeds.

QCD rejection at class-specific signal efficiencies

(a) Standard pair inputs

Signal	ϵ_s (%)	Shared	Layerwise	Shared + EV	Layerwise + EV
$H \to b\bar{b}$	50	5271 ± 242	5191 ± 195	5674 ± 152	5408 ± 248
$H \to c\bar{c}$	50	1741 ± 31	1792 ± 84	2007 ± 15	2020 ± 143
$H \to gg$	50	90.3 ± 0.8	91.2 ± 0.7	94.1 ± 0.8	93.7 ± 1.7
$H \to 4q$	50	665.0 ± 22.3	683.6 ± 32.5	813.5 ± 10.0	788.6 ± 24.0
$H \to q\bar{q}'\ell\nu$	99	1564 ± 23	1535 ± 126	1832 ± 32	1855 ± 106
$Z \to q\bar{q}$	50	229.8 ± 3.2	234.8 ± 5.0	249.5 ± 4.9	249.1 ± 4.4
$W \to q\bar{q}'$	50	284.7 ± 7.8	292.1 ± 2.7	315.2 ± 9.4	311.7 ± 2.3
$t \to bq\bar{q}'$	50	5833 ± 464	5629 ± 40	7395 ± 263	6932 ± 761
$t \to b\ell\nu$	99.5	3903 ± 161	3541 ± 348	4207 ± 182	4267 ± 378

QCD rejection at class-specific signal efficiencies

(b) Selected pair inputs (standard + PT + TRACK + REGION)

Signal	ϵ_s (%)	Shared	Layerwise	Shared + EV	Layerwise + EV
$H \to b\bar{b}$	50	4953 ± 147	5397 ± 88	5475 ± 368	5253 ± 316
$H \to c\bar{c}$	50	1907 ± 110	1976 ± 101	2128 ± 95	2234 ± 99
$H \to gg$	50	91.5 ± 1.1	93.3 ± 1.1	94.6 ± 1.7	95.4 ± 1.0
$H \to 4q$	50	748.5 ± 3.9	786.8 ± 22.6	817.9 ± 12.4	870.1 ± 12.7
$H \to q\bar{q}'\ell\nu$	99	1640 ± 86	1729 ± 146	1838 ± 80	1931 ± 28
$Z \to q\bar{q}$	50	239.6 ± 8.6	247.1 ± 1.2	252.5 ± 3.1	258.3 ± 6.6
$W \to q\bar{q}'$	50	303.7 ± 7.0	317.9 ± 6.3	323.8 ± 5.4	337.5 ± 8.4
$t \to bq\bar{q}'$	50	5787 ± 115	6682 ± 629	7282 ± 539	7997 ± 306
$t \to b\ell\nu$	99.5	3768 ± 278	3773 ± 216	4708 ± 268	4757 ± 314

Table 6: QCD background rejection at 30% signal efficiency on the official 20M-jet test sample for the seven hadronic classes. The two leptonic classes appear at their benchmark efficiencies in Table 5. Panels compare standard and selected pair inputs, with all models trained on 1M jets. Values are means ± sample standard deviations over three seeds.

QCD rejection at 30% signal efficiency

(a) Standard pair inputs

Signal	ϵ_s (%)	Shared	Layerwise	Shared + EV	Layerwise + EV
$H \to b\bar{b}$	30	17415 ± 784	17601 ± 1681	21410 ± 2118	18349 ± 2560
$H \to c\bar{c}$	30	7383 ± 372	8008 ± 783	8601 ± 250	8988 ± 1226
$H \to gg$	30	278.1 ± 4.6	281.8 ± 1.9	292.0 ± 8.7	290.0 ± 7.5
$H \to 4q$	30	2755 ± 25	2873 ± 181	3530 ± 219	3417 ± 106
$Z \to q\bar{q}$	30	1005 ± 41	1018 ± 36	1111 ± 25	1106 ± 50
$W \to q\bar{q}'$	30	1310 ± 68	1320 ± 16	1468 ± 82	1493 ± 75
$t \to bq\bar{q}'$	30	24442 ± 1392	26473 ± 2564	33139 ± 4066	29367 ± 4572

QCD rejection at 30% signal efficiency

(b) Selected pair inputs (standard + PT + TRACK + REGION)

Signal	ϵ_s (%)	Shared	Layerwise	Shared + EV	Layerwise + EV
$H \to b\bar{b}$	30	16684 ± 1236	18598 ± 779	19575 ± 1628	17602 ± 2514
$H \to c\bar{c}$	30	7958 ± 854	8475 ± 1121	9247 ± 906	9264 ± 953
$H \to gg$	30	280.5 ± 5.3	291.2 ± 4.9	292.5 ± 7.9	294.5 ± 5.1
$H \to 4q$	30	3192 ± 135	3460 ± 93	3643 ± 41	3937 ± 41
$Z \to q\bar{q}$	30	1054 ± 62	1090 ± 2	1124 ± 26	1173 ± 27
$W \to q\bar{q}'$	30	1429 ± 57	1494 ± 52	1527 ± 58	1626 ± 64
$t \to bq\bar{q}'$	30	25214 ± 370	28949 ± 2090	34017 ± 2417	38182 ± 5087